

Spring 2026 ASTR 4140/5140 Research Methods in Astrophysics

Project #2 -- Complete by Feb 12 (before class), 2026 (upload to Canvas).

In this project you will be working with SDSS data of the Sagittarius tidal stream in order to see if you can disentangle it from the Milky Way foreground/background. This project contains a number of intellectual steps that you need to work out. The corresponding coding/calculations should be relatively simple. Thus some background reading/literature search is encouraged while you try to work out what your approach should be. What you should upload to Canvas is a pdf file which concisely describes each step you took to see if you could characterize the Sagittarius stream, along with any figure/figures that describe this result (e.g., color-magnitude diagrams), and the very well-commented code that you used to make these plots. Please also include your SQL query.

The following paragraph gives you the background, and some potential motivation for this project, and the final two paragraphs describe the actual assignment.

The class lectures have covered various uses for tidal streams. In this assignment, you will see if you can recover evidence for the Sagittarius (Sgr) tidal stream in a new part of SDSS imaging. Specifically, you are targeting a field in the “trailing” part of the Stream (behind the Sgr dwarf galaxy in its orbit around the Milky Way, and this is exciting because this is a relatively unexplored part of the SDSS imaging footprint. A second reason that you might look to characterize the stream in existing imaging is if you want to write a proposal to get new imaging, say on the Hubble Space Telescope. Imagine, for instance, that you want to apply for time on HST in order to measure proper motions for these Sgr stream stars. To measure proper motions you need two epochs of data that are well spaced out in time, in order to measure relative motions between one epoch and the next. Given the extremely competitive nature of applying for and obtaining HST time, it will be hard to convince a TAC to guarantee you two epochs of time on the telescope, one now, and one several years from now. You therefore want to leverage data that already exists in the HST archive, if possible. You find that there is already very deep data of a field that is close to the track of the stream from Belokurov et al. 2006, at (RA,DEC) = (02:09:37.4, -04:38:31.6), taken with the HST/Advanced Camera for Surveys (ACS), in a broad I filter (called F814W), from circa 2003. You may be able to use this data as your first epoch! You want to estimate first, however, if there is indeed any evidence for the existence of Sgr stream stars in this region of sky by using the SDSS data. This in turn will inform whether your HST proposal is feasible.

This field contains stars related to the Sgr stream, but will also contain many more Milky Way stars in the foreground or background. To interpret this you want to use predictions of stellar population models. To identify stars in the color-magnitude diagram (CMD) that are consistent with belonging to the Sgr stream, you do a full literature search. You find that Sgr stream stars should fall into two different stellar populations: an old metal-poor population with (age, [Fe/H], $[\alpha/\text{Fe}]$) = (13 Gyr, -1.8, +0.2); and an intermediate metal-rich population (5 Gyr, -0.5, 0.0). The age and metallicity combinations were selected based on the HST CMD study of Siegel et al. (2007), and the $[\alpha/\text{Fe}]$ value for each metallicity was chosen based on the [Ti/Fe] versus [Fe/H] relation of Sgr stream stars studied by Chou et al. (2010). You estimate a distance from the ground-based data of Belokurov et al. (2006, 2014), obtaining 33.4 kpc for this field.

Your task now is to download SDSS data for this field, *fit* fiducial isochrones consistent with the findings above, and then ascertain whether you are seeing evidence for a Sgr-like population in this part of the sky. Since this project is about broad characterization, you can use a relatively large search area (like 1 x 1 degree). Keep in mind at what magnitude ranges you might trust the reported magnitudes for the survey, and then argue whether you are seeing a robust signal in this field. For fiducial isochrones, you can use those developed by the Dartmouth Stellar Evolution Database (Dotter et al. 2008: <http://stellar.dartmouth.edu/models/index.html>).

ASTR 5140 students: After reporting whether you are able to recover Sgr stream stars, describe one next step that you might take to characterize the stream better.

Several other papers that might be of use as you try to come up with a simple method to locate Sgr stream stars:

<https://ui.adsabs.harvard.edu/abs/2015ApJ...803...56S/abstract>

<https://ui.adsabs.harvard.edu/abs/2012ApJ...760L...6B/abstract>

Have fun!